

POWER BI

Q1. What is Power BI and why is it used in businesses?

Ans. Power BI is a Microsoft tool that turns raw data into visual charts and dashboards so businesses can understand their data easily and make better decision. Business use it because it shows all data into one place, creates real-time reports, and requires zero coding – just simple drag and drop.

Q2. Name and explain the three main components of Power BI

Ans.

1. Power BI Desktop It's a free app you install on your PC where you build reports and dashboards by connecting to data sources.
2. Power BI Service It's the online cloud version where you publish and share your reports with your team or organization.
3. Power BI Mobile It's the mobile app for iOS and Android where you can view your dashboards and reports anytime from your phone.

Q3. Explain the Power BI workflow.

Ans.

Step 1 — Connect & Get Data You connect Power BI to your data source like Excel, SQL, or any database and import the data.

Step 2 — Transform & Model You clean the data, remove errors, and create relationships between tables using Power Query.

Step 3 — Visualize & Share You build charts and dashboards in Power BI Desktop, then publish it to Power BI Service so others can view and access it.

Q4. List any four data cleaning tasks that can be performed in Power Query.

Ans.

1. Removing Duplicates Delete repeated rows so data stays unique and accurate.
2. Handling Missing Values Fill empty cells with a default value or remove those rows completely.
3. Changing Data Types Convert text to numbers, numbers to dates etc. so data is in the correct format.
4. Renaming Columns Give proper meaningful names to columns instead of confusing default names.

Q6. Define Data View, Report View, and Model View. Explain the purpose of each view.

Ans.

1. Data View

Definition:

Data View is used to view the dataset in rows and columns format.

Purpose:

- To check data properly
- To find missing values
- To understand the dataset structure

2. Report View

Definition:

Report View is the main workspace where charts and reports are created.

Purpose:

- To create visualizations
- To design dashboards
- To analyze data visually

3. Model View

Definition:

Model View shows relationships between different tables.

Purpose:

- To connect tables
- To manage relationships
- To build a proper data model

Q7. Discuss the different data sources that Power BI supports.

Ans.

1. Excel Files

Power BI can import data from Excel spreadsheets.

2. CSV/Text Files

CSV and text files can be loaded directly into Power BI.

3. SQL Server

Power BI supports SQL Server databases for data analysis.

4. Web Sources

Data can be imported from websites and APIs.

5. Cloud Services

Power BI supports cloud services like:

- Azure
- SharePoint
- Google Analytics

6. Oracle Database

Power BI can connect with Oracle databases.

7. MySQL Database

Power BI also supports MySQL databases for importing data.